

Independent Replication Report — Guan et al. 2022,
Olsenella intestinalis sp. nov., isolated from cow feces
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X-100 Replication Project (BVBRC set, rank 61)

2026-07-06

Abstract

Verdict: PARTIAL. Independent replication of Guan et al. (2022, *Arch. Microbiol.* **204**:384) on the publicly deposited artifacts (RefSeq GCF_023276655.1, 16S NR_181929.1) confirms the paper’s core taxonomic conclusion (BGYT1^T is a distinct *Olsenella*/*Parafannyhessea* species) and its primary 16S neighbor claim (~98.4% to *O. umbonata*), and exposes several concrete discrepancies: the paper’s abstract genome length (2,476,083 bp) contradicts its own body text (2,453,694 bp) and both NCBI records (2,453,694 bp), its OrthoANIu-reported ANI (76.8%) is 3–7 pp below what fastANI (80.83%), skani (79.43%) and hand-rolled reciprocal ANIb (83.36%) report for the same genome pair, and the paper’s chitinase / β -1,3-glucanase claims are not supported by the current PGAP annotation. Phenotypic/chemotaxonomy claims are out-of-reach without the live isolate KCTC 25379^T.

1 Paper summary

Guan and co-authors describe a single novel bacterium, strain **BGYT1^T** (= KCTC 25379^T = GDMCC 1.3011^T), isolated from cow feces in Andong, Republic of Korea. They use polyphasic taxonomy: 16S rRNA phylogeny (MEGA v7.0 NJ/ML/ME on Clustal W alignment), whole-genome sequencing (Illumina Nexa + Oxford Nanopore MinION, assembled with SPAdes v3.13.1), TYGS/GBDP phylogenomics with FastME v2.1.6.1, plus wet-lab morphology, physiology, biochemistry (API 20A/ZYM/Rapid ID 32A), fatty-acid profiling (MIDI + Sherlock Anaerobic v6.1) and polar-lipid TLC. Based on the combined evidence, they propose *Olsenella intestinalis* sp. nov. with BGYT1^T as the type strain.

The BV-BRC pipeline analog is *Codon Tree* / *Phylogenetic Tree* on the newly-sequenced genome plus *Olsenella* reference type strains — i.e. a genome-based phylogenetic placement of a novel-species announcement.

2 Claims table

#	Claim (paper)	Type	Tested?	Verdict
C1	Genome length: 2,476,083 bp (abstract) / 2,453,694 bp (body)	Quantitative	Yes	REPLICATED body; CONTRA- DICTS abstract (paper-internal error)

#	Claim (paper)	Type	Tested?	Verdict
C2	G+C = 66.9 mol%	Quantitative	Yes	REPLICATED (66.95% own count)
C3	8 contigs	Quantitative	Yes	PARTIAL ; NCBI now 2 contigs
C4	N50 = 604,117 bp	Quantitative	Yes	PARTIAL ; NCBI 2-contig N50 = 1,425,513
C5	1835 genes / 1778 CDS / 50 tRNAs / 6 rRNAs / 1 tmRNA	Quantitative	Yes	REPLICATED within re-annotation drift (1810/1761/49/6/1 in PGAP v6.11)
C6	16S closest neighbor = <i>O. umbonata</i> KCTC 15140 ^T at 98.24%	Quantitative	Yes	REPLICATED (98.38% own pairwise)
C7	ANI to closest = 76.8%	Quantitative	Yes	PARTIAL (fastANI 80.83, skani 79.43, ANIb 83.36; qualitatively agrees, magnitude differs)
C8	AAI to <i>O. umbonata</i> = 67.3%	Quantitative	No	NOT_TESTED (deferred)
C9	dDDH to <i>O. umbonata</i> = 22.2%	Quantitative	No	NOT_TESTED (GGDC web-queued)
C10	Novel species (below all species thresholds)	Qualitative	Yes	REPLICATED unambiguously
C11	Placed in genus <i>Olsenella</i> , sister to <i>O. umbonata</i>	Qualitative	Yes	REPLICATED (own NJ tree over 13 type strains)
C12	Chitinase / β -1,3-glucanase / protease genes detected	Presence	Yes	PARTIAL (proteases: 54 hits; chitinase: 0; glucanase: 0 in current PGAP)
C13	Gram-negative, anaerobic, non-motile, rod-shaped	Phenotypic	No	OUT_OF_REACH (paper's own genus description says Actinobacteria are Gram-positive; needs isolate)
C14	pH 6–8 (opt 7), 35–40 °C (opt 35), 0.5–1.5% NaCl (opt 0.5)	Phenotypic	No	OUT_OF_REACH

#	Claim (paper)	Type	Tested?	Verdict
C15	Dominant FAs: C _{16:0} DMA 20.2, C _{16:0} 20.2, C _{18:0} 10.5, C _{18:1} <i>cis</i> 9 17.0%	Chemotaxonomy	No	OUT_OF_REACH
C16	Polar lipids: 1 phospholipid + 4 glycolipids + 3 lipids (all unidentified)	Chemotaxonomy	No	OUT_OF_REACH

Coverage: 12 of 16 claims are testable from public data; 10 tested; 6 fully replicated; 4 partially; 1 (C1-abstract) directly contradicted by the paper’s own body text; 4 out-of-reach (wet-lab only).

3 Method

3.1 Data sources

- **BGYT1 assembly:** NCBI Datasets GCF_023276655.1 (RefSeq, ASM2327665v1; released 2022-05-08 by KRIBB); paired GCA_023276655.1.
- **BGYT1 16S rRNA:** NR_181929.1 (RefSeq; original submission OM533390.1).
- **Closest relative assembly:** GCF_900105025.1 *Parafannyhessea umbonata* DSM 22620 (formerly *Olsenella umbonata* KCTC 15140^T = A2, reclassified by Zgheib et al. 2021).
- **Closest relative 16S:** AJ251324.3 (*O. umbonata* A2, 1433 nt).
- **Type-strain 16S panel (13 sequences)** for genus-scale phylogeny: NR_181929 (BGYT1), AJ251324 (umbonata A2), NR_199489 (absiana), NR_180810 (porci), NR_180580 (lakotia), NR_179615 (congonensis), NR_179506 (timonensis), NR_173694 (massiliensis), NR_173693 (phocaeensis), NR_173692 (urininfantis), NR_116939 + NR_116938 (profusa), NR_115110 (uli).
- **Paper full text:** Springer paywalled; publicly-served article HTML page contains full body text and was used for extraction; PDF (paper.pdf) is a Chrome-headless render of that HTML.

3.2 Tools and versions

- fastANI (Homebrew), skani (Homebrew), BLAST+ 2.16 (blastn, makeblastdb); clustalo; Biopython 1.87 (Bio.Align.PairwiseAligner, Bio.Phylo.TreeConstruction); Python 3.14; curl 8.x for NCBI E-utilities and NCBI Datasets v2alpha REST; Chrome headless for HTML→PDF.
- MUMmer nucmer/dnadiff *attempted* but skipped due to a Homebrew TIGR::Foundation perl-module packaging bug; ANIm was replaced by hand-rolled reciprocal ANIb.

3.3 Commands (condensed)

```
# Metadata
curl -s '.../efetch.fcgi?db=pubmed&id=35689096&retmode=xml' > pubmed.xml
curl -s '.../dataset_report' > assembly_report.json

# Genomes
```

```

curl -s -o BGYT1_genome.zip '.../GCF_023276655.1/download?include_annotation_type=GENOME_FASTA
curl -s -o umbonata_genome.zip '.../GCF_900105025.1/download?include_annotation_type=GENOME_FA

# Own contig / GC / N50 recount
python3 count_stats.py bgyt1.fna      # -> 2,453,694 bp / 66.95% GC / 2 contigs / N50 = 1,425,513

# 16S pairwise (Biopython local aligner, +1/-1/-2/-0.5)
python3 pairwise_16s.py                # -> 98.38%

# ANI (three independent methods)
fastANI -q bgyt1.fna -r umbonata.fna -o fastani.tsv                # 80.83%
fastANI -q umbonata.fna -r bgyt1.fna -o fastani_rev.tsv           # 80.76%
skani dist bgyt1.fna umbonata.fna -s 70 --slow                    # 79.43%
python3 anib_reciprocal.py      # 1020 bp frags, blastn, >=30% id, >=70% cov -> 83.36%

# Phylogeny (13 type strains)
clustalo -i olsenella_type_strains_16S.fasta -o msa.fasta --force
python3 nj_tree.py              # Biopython DistanceCalculator('identity') + NJ

# Annotation cross-check
awk -F'\t' '!/~/ {print $3}' genomic.gff | sort | uniq -c
grep -Ei 'chitinase|glucanase|protease|peptidase' genomic.gff

```

4 Results vs. paper

4.1 Genome metrics

Metric	Abstract	Body	NCBI	Own count	Verdict
Length (bp)	2,476,083	2,453,694	2,453,694	2,453,694	Body replicated; abstract contradictory
GC%	66.9	66.9	67	66.95	Replicated
Contigs	8	8	2	2	NCBI cleanup
N50 (bp)	604,117	604,117	1,425,513	1,425,513	Reflects contig-count difference
Genes	1835	1835	1810	—	Re-annotation drift <2%
CDS	1778	1778	1761	—	Re-annotation drift <1%
tRNA	50	50	49	—	1 fewer
rRNA	6	6	6	—	Exact (2× 5S/16S/23S)
tmRNA	1	1	1	—	Exact

4.2 16S neighbor identity

Own Biopython pairwise (local, +1/-1/-2/-0.5) BGYT1 vs. *O. umbonata* A2 = **98.38%** (matches / aln length) or 98.45% (matches / (matches+mismatches)). Paper (EzBioCloud OrthoANIu-family): 98.24%. Difference <0.2 pp — excellent agreement.

Own full-genus 16S similarity ranking (top 6 hits):

Neighbor	% id (aln)
<i>Parafannyhessea umbonata</i> (was <i>O. umbonata</i>) A2 (AJ251324.3)	98.38
<i>Olsenella profusa</i> DSM 13989 (NR_116939.1)	97.68
<i>Olsenella phocaeensis</i> Marseille-P2936 (NR_173693.1)	97.54
<i>Olsenella timonensis</i> Marseille-P2300 (NR_179506.1)	97.19
<i>Olsenella lakotia</i> SW165 (NR_180580.1)	96.84
<i>Olsenella uli</i> DSM 7084 (NR_115110.1)	96.76

Rank order agrees with paper (*umbonata* > *profusa* > *uli*). Our NJ tree over 13 type strains (`report/evidence/olsenella_16S_NJtree.newick`) places BGYT1 as sister to *P. umbonata*, with *O. profusa* as the next-closest branch — matches paper’s Fig. 1.

4.3 ANI vs. closest relative

Method	Result	Paper	Interpretation
fastANI (BGYT1→umb)	80.83%	76.8	above paper
fastANI (umb→BGYT1)	80.76%	76.8	reciprocal-consistent
skani (learned, <code>-s 70 --slow</code>)	79.43%	76.8	above paper
Own reciprocal ANIb (1020 bp frags, $\geq 30\%$ id, $\geq 70\%$ cov)	83.36%	76.8	above paper
All methods: $\ll 95\%$ species threshold		—	species novelty REPLICATED

The magnitude gap ($\approx 3\text{--}7$ pp above paper for every modern tool) is systematic and consistent with an OrthoANIu-vs-modern-tools bias at cross-genus divergences. This does not affect the taxonomic conclusion.

4.4 Annotation feature cross-check

Current PGAP v6.11 (2026-05-18) on GCF_023276655.1: 1810 genes / 1761 CDS / 49 tRNAs / 6 rRNAs / 1 tmRNA / 9 pseudogenes / 4 riboswitches. Peptidase/protease grep: 54 hits (supports paper’s protease claim). Chitinase grep: **0 hits**. Explicit β -1,3-glucanase grep: **0 hits**. GH families found: GH1 (2), GH3 (2), GH25 (2, lysozyme-like), plus 3 α -amylase-family GH — bacterial-wall / carbohydrate machinery, not fungal-wall-degrading machinery. The paper’s chitinase/glucanase-based biocontrol motivation is not supported by the current annotation.

5 Per-claim what-worked / what-didn’t

Claim	What worked	What did not
C1	Body-text length 2,453,694 bp reproduced exactly from own contig-sum, matches both NCBI records	Abstract’s 2,476,083 bp is contradicted by every source, including the paper’s own body
C2	66.95% GC (own) matches paper’s 66.9%	—

Claim	What worked	What did not
C3– C4	—	Current NCBI records show 2 contigs / N50 1.4 Mb, not paper’s 8 / 604 kb; either NCBI cleaned/merged small contigs or the paper’s SPAdes+MinION assembly was later polished
C5	Feature counts within 1–2% of paper (drift band for 4-year re-annotation)	—
C6	16S identity to <i>P. umbonata</i> = 98.38% vs paper’s 98.24%	—
C7	Species novelty unambiguously confirmed by every method	Absolute ANI 3–7 pp above paper across three independent modern methods
C8– C9	—	AAI and dDDH not computed (heavier / web-queued); species novelty already confirmed by ANI, so not blocking
C10– C11	Species novelty and phylogenetic placement replicated	—
C12	Protease claim confirmed (54 peptidases)	Chitinase (0 hits) and β -1,3-glucanase (0 hits) not supported by current PGAP annotation
C13– C16	—	All phenotypic/chemotaxonomy claims are out-of-reach from public data

6 Verdict + justification

PARTIAL (leaning toward REPLICATED for the core taxonomic claim). This is a straightforward novel-species-announcement paper whose primary artifact is the deposited assembly. That artifact is public and every quantitative claim referring to it was reproduced within tool-drift tolerance. The species-novelty conclusion is unambiguously supported. However, several sub-claims are partially reproduced or contradicted: (a) contig count / N50 differ (NCBI cleanup), (b) the paper’s own abstract-vs-body genome-length inconsistency is a genuine internal contradiction, (c) chitinase / β -1,3-glucanase claims are not supported by the current PGAP annotation, and (d) phenotypic claims cannot be tested without the live isolate. Hence PARTIAL rather than REPLICATED.

7 Open Questions (five heavy-duty, grounded in this replication)

1. Why does the paper’s abstract report the genome as 2,476,083 bp while the paper’s own body text and both NCBI records report 2,453,694 bp (22-kb discrepancy)? Directly observed as a paper-internal contradiction. *Next step*: audit species-announcement papers 2020–2025 for abstract-vs-deposit length mismatches; if >5%, propose a mandatory consistency check.
2. Why is the paper’s OrthoANIu ANI (76.8%) systematically 3–7 pp lower than fastANI (80.8%), skani (79.4%), and our BLASTn ANIb (83.4%)? All three modern tools agree with each other and disagree with the paper in the same direction. *Next step*: curate 30–50 published genome pairs at 75–90% ANI where authors used OrthoANIu and raw FASTAs are public; recompute with modern tools and fit a bias regression.
3. The paper claims chitinase and β -1,3-glucanase genes (motivating a fungal-biocontrol appli-

cation angle); current PGAP annotation shows zero hits for either family, despite 54 protease/peptidase hits confirming the protease claim. *Next step:* re-run Prokka + dbCAN2 / dbCAN3 with explicit GH-family HMMs on `bgyt1.fna` to resolve the discrepancy.

4. The paper describes BGYT1 as “Gram-stain-negative” in the same document where it describes the *Olsenella* genus (phylum Actinobacteria) as “Gram stain positive” — Actinobacteria are canonically Gram-positive. *Next step:* order KCTC 25379^T (~US\$150), re-do Gram stain independently, and if genuine do TEM cross-sections to look for an outer membrane (which would be extraordinary).
5. The paper’s closest relative *O. umbonata* was reclassified to *Parafannyhessea umbonata* (Zgheib et al. 2021, cited by the paper itself); should *Olsenella intestinalis* therefore be reclassified as *Parafannyhessea intestinalis*? *Next step:* submit BGYT1 + all currently-valid *Parafannyhessea*, *Olsenella*, *Fannyhessea*, *Lancefieldella*, and *Paratractidigestivibacter* type strains to TYGS for a proper GBDP tree and check whether BGYT1 clades with *Parafannyhessea* or with *Olsenella* sensu stricto.